

# DDL, DCL, DML, TCL Commands

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## Create a database & use :

```
1 CREATE DATABASE IF NOT EXISTS bookflow2_db;
2 USE bookflow_db;
```

Output

| # | Time     | Action                                     |
|---|----------|--------------------------------------------|
| 1 | 09:08:46 | SHOW DATABASES                             |
| 2 | 09:09:23 | CREATE DATABASE IF NOT EXISTS bookflow2_db |
| 3 | 09:09:40 | USE bookflow_db                            |

## Create a table

```
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
11
```

Output

| # | Time     | Action                                                         | Message           |
|---|----------|----------------------------------------------------------------|-------------------|
| 1 | 09:08:46 | SHOW DATABASES                                                 | 4 row(s) returned |
| 2 | 09:09:23 | CREATE DATABASE IF NOT EXISTS bookflow2_db                     | 1 row(s) affected |
| 3 | 09:09:40 | USE bookflow_db                                                | 0 row(s) affected |
| 4 | 09:12:49 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, cu... | 0 row(s) affected |

## Alter table add column

```
11 ALTER TABLE bank_transactions
12 ADD account_no VARCHAR(20);
```

Output

| # | Time     | Action                                                         |
|---|----------|----------------------------------------------------------------|
| 1 | 09:08:46 | SHOW DATABASES                                                 |
| 2 | 09:09:23 | CREATE DATABASE IF NOT EXISTS bookflow2_db                     |
| 3 | 09:09:40 | USE bookflow_db                                                |
| 4 | 09:12:49 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, cu... |
| 5 | 09:13:57 | ALTER TABLE bank_transactions ADD account_no VARCHAR(20)       |

## Alter table modify column

```
13 • ALTER TABLE bank_transactions
14 MODIFY customer_name VARCHAR(100);
```

Output

Action Output

| #   | Time     | Action                                                          |
|-----|----------|-----------------------------------------------------------------|
| ✓ 1 | 09:08:46 | SHOW DATABASES                                                  |
| ✓ 2 | 09:09:23 | CREATE DATABASE IF NOT EXISTS bookflow2_db                      |
| ✓ 3 | 09:09:40 | USE bookflow_db                                                 |
| ✓ 4 | 09:12:49 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, cu...  |
| ✓ 5 | 09:13:57 | ALTER TABLE bank_transactions ADD account_no VARCHAR(20)        |
| ✓ 6 | 09:15:57 | ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(1... |

## Rename table

```
17 • SELECT * FROM bank_transactions;
18 • RENAME TABLE bank_transactions
19 TO customer_transactions;
20
```

Output

Action Output

| #    | Time     | Action                                                  |
|------|----------|---------------------------------------------------------|
| ✓ 9  | 09:18:07 | SHOW TABLES                                             |
| ✓ 10 | 09:20:03 | RENAME TABLE bank_transactions TO customer_transactions |

## Truncate table:

```
20 • TRUNCATE TABLE customer_transactions;
21
```

Output

Action Output

| #    | Time     | Action                               |
|------|----------|--------------------------------------|
| ✗ 11 | 09:22:08 | TRUNCATE TABLE bank_transactions     |
| ✓ 12 | 09:22:27 | TRUNCATE TABLE customer_transactions |

## Drop table:

```
21 • DROP TABLE IF EXISTS bank_backup;
22
```

Output

Action Output

| #    | Time     | Action                               |
|------|----------|--------------------------------------|
| ✓ 12 | 09:22:27 | TRUNCATE TABLE customer_transactions |
| ⚠ 13 | 09:23:21 | DROP TABLE IF EXISTS bank_backup     |

## DML (DATA MANIPULATION LANGUAGE)

### Create table:

```

22 • CREATE TABLE bank_transactions (
23     txn_id INT PRIMARY KEY,
24     customer_name VARCHAR(50),
25     branch_name VARCHAR(50),
26     transaction_type VARCHAR(20),
27     amount DECIMAL(10,2),
28     transaction_date DATE
29 );

```

Output



Action Output

| #  | Time     | Action                                                   |
|----|----------|----------------------------------------------------------|
| 13 | 09:23:21 | DROP TABLE IF EXISTS bank_backup                         |
| 14 | 09:24:21 | CREATE TABLE bank_transactions ( txn_id INT PRIMARY KEY, |

## Insert sample data:

```

31 • INSERT INTO bank_transactions VALUES
32     (101, 'Likhitha', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
33     (102, 'Tejaswini', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
34     (103, 'Teju', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
35     (104, 'Krishna', 'Vizag', 'Deposit', 8000, '2024-01-10'),
36     (105, 'Shiva', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
37     (106, 'Navya', 'Vizag', 'Deposit', 15000, '2024-01-12'),
38     (107, 'Kavya', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
39     (108, 'Akshitha', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
40     (109, 'Lakshmi', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
41     (110, 'Saraswati', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
42 • SELECT * FROM bank_transactions;

```

|   | txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|---|--------|---------------|-------------|------------------|----------|------------------|
| ▶ | 101    | Likhitha      | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
|   | 102    | Tejaswini     | Hyderabad   | Withdrawal       | 2000.00  | 2024-01-06       |
|   | 103    | Teju          | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|   | 104    | Krishna       | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
|   | 105    | Shiva         | Hyderabad   | Withdrawal       | 3500.00  | 2024-01-11       |
|   | 106    | Navya         | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|   | 107    | Kavya         | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       |
|   | 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
|   | 109    | Lakshmi       | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       |
|   | 110    | Saraswati     | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|   | NULL   | NULL          | NULL        | NULL             | NULL     | NULL             |

## Insert

```

45  (111,'Pranavi','Vizag','Deposit',7000,'2024-01-18');
46  • SELECT * FROM bank_transactions;

```

| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|--------|---------------|-------------|------------------|----------|------------------|
| 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
| 109    | Lakshmi       | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       |
| 110    | Saraswati     | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
| 111    | Pranavi       | Vizag       | Deposit          | 7000.00  | 2024-01-18       |

## Update:

```

47  • UPDATE bank_transactions
48  SET amount = 3000
49  WHERE txn_id = 105;
50  • SELECT * FROM bank_transactions;

```

| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|--------|---------------|-------------|------------------|----------|------------------|
| 104    | Krishna       | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
| 105    | Shiva         | Hyderabad   | Withdrawal       | 3000.00  | 2024-01-11       |
| 106    | Navya         | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
| 107    | Kavya         | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       |
| 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |

## Delete and select:

```

51  • DELETE FROM bank_transactions
52  WHERE txn_id = 111;
53  • SELECT * FROM bank_transactions;

```

| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|--------|---------------|-------------|------------------|----------|------------------|
| 107    | Kavya         | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       |
| 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
| 109    | Lakshmi       | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       |
| 110    | Saraswati     | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
| NULL   | NULL          | NULL        | NULL             | NULL     | NULL             |

## Select with where:

```

54  • SELECT *
55  FROM bank_transactions
56  WHERE transaction_type='Deposit';

```

| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|--------|---------------|-------------|------------------|----------|------------------|
| 101    | Likitha       | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
| 103    | Teju          | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
| 104    | Krishna       | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
| 106    | Navya         | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
| 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
| 110    | Saraswati     | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |

## Select with order by:

```

57 • SELECT *
58 FROM bank_transactions
59 ORDER BY amount DESC;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

|   | txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
|---|--------|---------------|-------------|------------------|----------|------------------|
| ▶ | 106    | Navya         | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|   | 103    | Teju          | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|   | 110    | Saraswati     | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|   | 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
|   | 104    | Krishna       | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
|   | 101    | Likhitha      | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
|   | 109    | Lakshmi       | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       |
|   | 105    | Shiva         | Hyderabad   | Withdrawal       | 3000.00  | 2024-01-11       |
|   | 102    | Tejaswini     | Hyderabad   | Withdrawal       | 2000.00  | 2024-01-06       |
|   | 107    | Kavya         | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       |
| ✱ | NULL   | NULL          | NULL        | NULL             | NULL     | NULL             |

## DCL (DATA CONTROL LANGUAGE)

### GRANT ALL PRIVILEGES:

```

60 • CREATE USER 'BranchManager'@'localhost'
61 IDENTIFIED BY 'Branch@123';
62 • GRANT ALL PRIVILEGES
63 ON bookflow1_db.bank_transactions
64 TO 'BranchManager'@'localhost';
65 • SHOW GRANTS FOR 'BranchManager'@'localhost';

```

| Result Grid                                      | Filter Rows: | Export: | Wrap Cell Content |
|--------------------------------------------------|--------------|---------|-------------------|
| Grants for BranchManager@localhost               |              |         |                   |
| GRANT USAGE ON *,* TO 'BranchManager'@'...'...   |              |         |                   |
| GRANT ALL PRIVILEGES ON 'bookflow1_db'.'b...'... |              |         |                   |

### Revoke all:

```

66 • REVOKE ALL PRIVILEGES
67 ON bookflow1_db.bank_transactions
68 FROM 'BranchManager'@'localhost';

```

Output



Action Output

| #  | Time     | Action                                                           |
|----|----------|------------------------------------------------------------------|
| 34 | 09:37:33 | CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'Branch... |
| 35 | 09:37:35 | GRANT ALL PRIVILEGES ON bookflow1_db.bank_transactions TO 'Br... |
| 36 | 09:37:37 | SHOW GRANTS FOR 'BranchManager'@'localhost'                      |
| 37 | 09:38:45 | REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FRO...   |

# TCL (TRANSACTION CONTROL LANGUAGES)

## Commit:

```
69 • UPDATE bank_transactions
70 SET amount=6000
71 WHERE txn_id=101;
72 • COMMIT;
```

Output

Action Output

| #    | Time     | Action                                                         |
|------|----------|----------------------------------------------------------------|
| ✓ 36 | 09:37:37 | SHOW GRANTS FOR 'BranchManager'@'localhost'                    |
| ✓ 37 | 09:38:45 | REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FRO... |
| ✓ 38 | 09:39:50 | UPDATE bank_transactions SET amount=6000 WHERE txn_id=101      |
| ✓ 39 | 09:39:52 | COMMIT                                                         |

## Savepoint:

```
73 • SAVEPOINT Before_Update;
```

Output

Action Output

| #    | Time     | Action                                                       |
|------|----------|--------------------------------------------------------------|
| ✓ 37 | 09:38:45 | REVOKE ALL PRIVILEGES ON bookflow1_db.bank_transactions FRO. |
| ✓ 38 | 09:39:50 | UPDATE bank_transactions SET amount=6000 WHERE txn_id=101    |
| ✓ 39 | 09:39:52 | COMMIT                                                       |
| ✓ 40 | 09:40:55 | SAVEPOINT Before_Update                                      |

## Rollback to savepoint:

```
75 SET amount=99999
76 WHERE txn_id=102;
77 • SELECT * FROM bank_transactions;
```

| Result Grid |        | Filter Rows:  |             | Edit:            | Export/Import: |                  |
|-------------|--------|---------------|-------------|------------------|----------------|------------------|
|             | txn_id | customer_name | branch_name | transaction_type | amount         | transaction_date |
|             | 101    | Likhitha      | Hyderabad   | Deposit          | 6000.00        | 2024-01-05       |
|             | 102    | Tejaswini     | Hyderabad   | Withdrawal       | 99999.00       | 2024-01-06       |
|             | 103    | Teju          | Vijayawada  | Deposit          | 12000.00       | 2024-01-08       |
|             | 104    | Krishna       | Vizag       | Deposit          | 8000.00        | 2024-01-10       |
|             | 105    | Shiva         | Hyderabad   | Withdrawal       | 3000.00        | 2024-01-11       |
|             | 106    | Navya         | Vizag       | Deposit          | 15000.00       | 2024-01-12       |
|             | 107    | Kavya         | Vijayawada  | Withdrawal       | 1000.00        | 2024-01-13       |
|             | 108    | Akshitha      | Hyderabad   | Deposit          | 9000.00        | 2024-01-14       |
|             | 109    | Lakshmi       | Vizag       | Withdrawal       | 4000.00        | 2024-01-15       |
|             | 110    | Saraswati     | Vijayawada  | Deposit          | 11000.00       | 2024-01-16       |
|             | NULL   | NULL          | NULL        | NULL             | NULL           | NULL             |